Week 46: Two boards forget their starts

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Problems and routes

Problem	K–1	2–3	4–5	route
The fewest instructions to match	P1	P1	P1	first
Fewest instructions for four pairs	P2			first
Invent a pair needing two instructions	P3			return
Which stories guarantee a match?		P2		first
Matching before every slot is marked		P3		first
How many final boards can a story leave?			P2	first
When a story forgets every start			P3	first
Two instructions: a match and a miss	P4			first
Three instructions that always match	P5			return
Six random draws	P6			first
After the boards match	P7			first
Must three draws make a match?		P4		return
The shortest story that always matches		P5		return
Can shared toggles make boards match?		P6	P4	return
Stories with setting instructions and toggles		P7	P5	return
Slots 1, 3, 1, 2: which final boards are likeliest?			P6	return
Any slot story that visits all three slots			P7	return

First route. K–1 P1–P2, P4, P6–P7; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P3, P5; Grades 2–3 P4–P7; Grades 4–5 P4–P7.

Key, one entry per problem

The fewest instructions to match

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Two boards can be made to match with exactly as many instructions as slots where they differ: each differing slot needs one, and one per slot suffices.

Fewest instructions for four pairs

K–1 P2 · first route

Idea. RBR/RRR needs 1, RRB/BRR 2, BRB/BRB 0, RRR/BBB 3: the number of differing slots.

Invent a pair needing two instructions

K–1 P3 · return route

Idea. Any pair differing in exactly two slots needs two; no pair needs four, since there are only three slots.

Which stories guarantee a match?

Grades 2–3 P2 · first route

Idea. A fixed story makes every pair of starts match exactly when it sets every slot: only 2->R, 3->B, 1->R does.

Matching before every slot is marked

Grades 2–3 P3 · first route

Idea. Two different boards match as soon as every slot where they differed has been set; slots that already agreed need no mark.

How many final boards can a story leave?

Grades 4–5 P2 · first route

Idea. Each set slot ends at its last assigned color; with u untouched slots, 2^u final boards remain from the eight starts.

When a story forgets every start

Grades 4–5 P3 · first route

Idea. A setting story forgets every start exactly when it visits all three slots; two particular boards can agree sooner if they agreed on every untouched slot.

Two instructions: a match and a miss

K–1 P4 · first route

Idea. After 1->R and 3->B only the middle color survives, so two different starts end matching exactly when their middle colors agree.

Three instructions that always match

K–1 P5 · return route

Idea. One instruction per slot makes every start finish the same; two instructions leave a slot unset, where two starts can still differ.

Six random draws

K–1 P6 · first route

Idea. Random draws can repeat a slot, so matching can take more than three draws.

After the boards match

K–1 P7 · first route

Idea. A shared instruction puts the same color in the same slot of both boards, so matching boards stay matched.

Must three draws make a match?

Grades 2–3 P4 · return route

Idea. Draws can repeat a slot (1->R three or ten times), so no fixed number of draws guarantees a match.

The shortest story that always matches

Grades 2–3 P5 · return route

Idea. Three instructions, one per slot; any shorter story misses a slot where two starts can differ.

Can shared toggles make boards match?

Grades 2–3 P6 · Grades 4–5 P4 · return route · wording differs by band

Idea. A shared toggle keeps every disagreement a disagreement, so toggles alone never merge two different boards.

Stories with setting instructions and toggles

Grades 2–3 P7 · Grades 4–5 P5 · return route · wording differs by band

Idea. A mixed story makes every start finish the same exactly when each slot gets at least one color-setting instruction.

Slots 1, 3, 1, 2: which final boards are likeliest?

Grades 4–5 P6 · return route

Idea. The final board is the last colors set at slots 1, 2, 3, so each of the eight boards comes from exactly 2 of the 16 color stories: all tie at 1/8.

Any slot story that visits all three slots

Grades 4–5 P7 · return route

Idea. Any fixed slot story visiting every slot gives all eight boards chance 1/8; four random draws need not visit every slot.